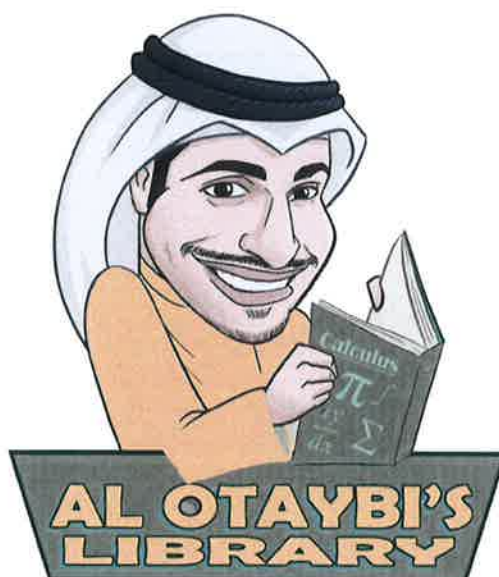


Kuwait University

2021/2022

Spring course



Calculus B

Final

Sections : **10.1 – 10.2**

10.1 Curves Defined by Parametric Equations.

10.2 Calculus with Parametric Curves.

(16)

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10.1 – 10.2 Parametric curves



$$x = t \dots$$
$$y = t \dots$$

Convert a parametric
equation to
x and y equation

Arc length

Derivatives

Sketch

Area under the
curve

Surface Area

Centroid

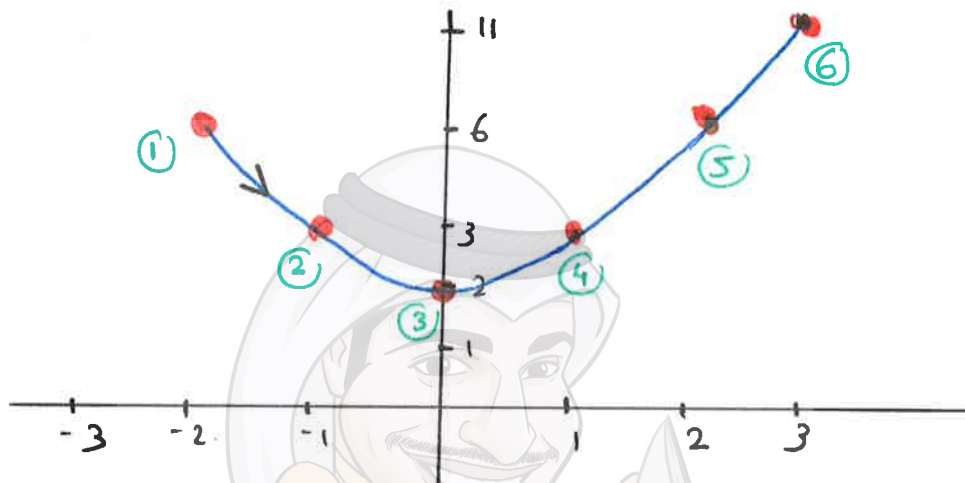


Find the equation in x and y and indicate the orientation of the parametric curve such that

$$x = t, \quad y = t^2 + 2, \quad -2 \leq t \leq 3$$

$$y = x^2 + 2$$

t	-2	-1	0	1	2	3
x	-2	-1	0	1	2	3
y	6	3	2	3	6	11



Find the equation in x and y and indicate the orientation of the parametric curve such that

$$x = t^2 - 2t, \quad y = t + 1, \quad t \in \mathbb{R}$$

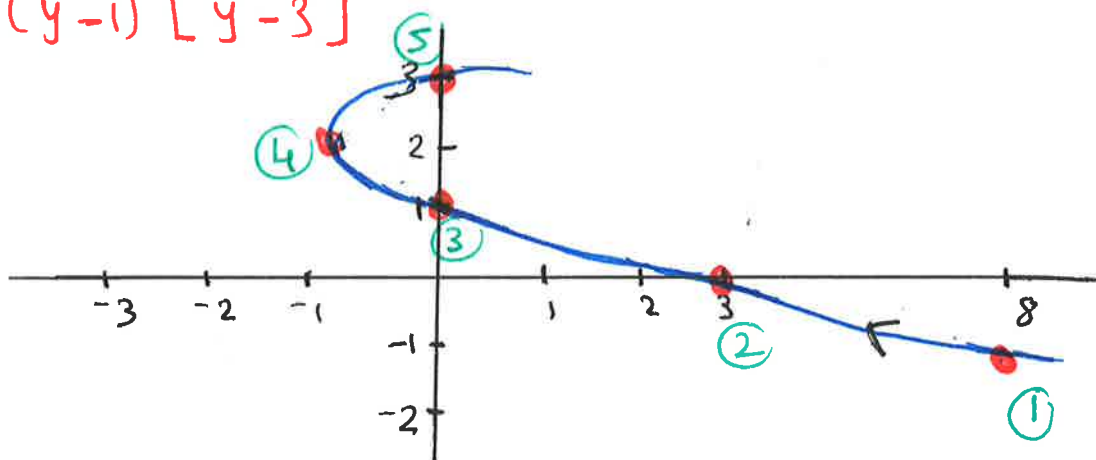
$$x = t^2 - 2t \quad t = y - 1$$

$$x = (y - 1)^2 - 2(y - 1)$$

$$x = (y - 1)[y - 1 - 2]$$

$$x = (y - 1)[y - 3]$$

t	-2	-1	0	1	2
x	8	3	0	-1	0
y	-1	0	1	2	3



Find the equation in x and y and sketch the parametric curve

$$x = 1 + 2t, \quad y = t - 1, \quad 0 \leq t \leq 1$$

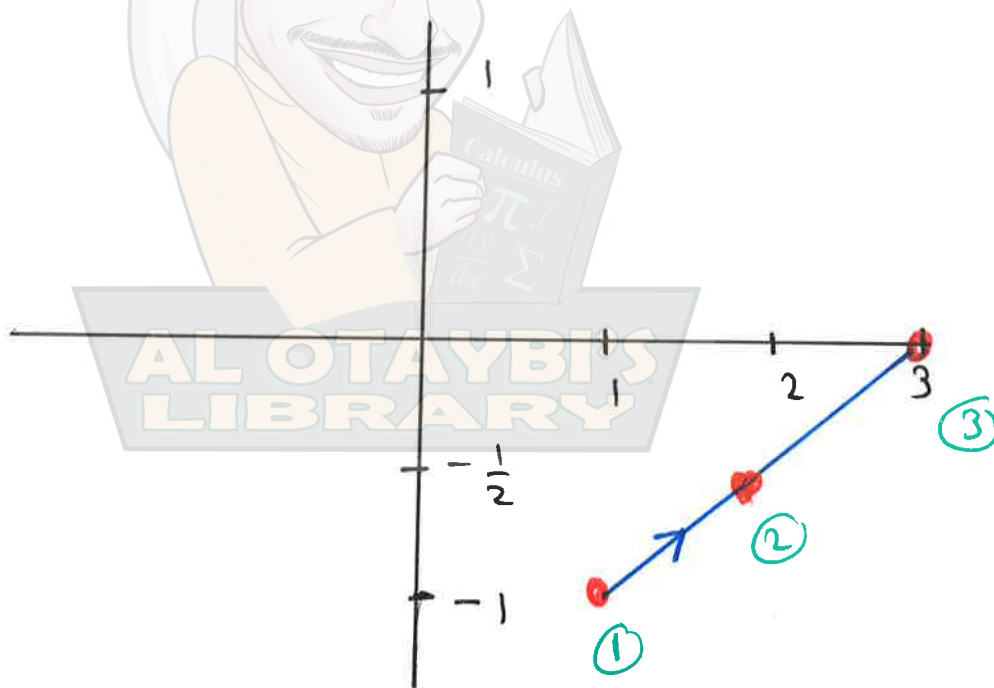
$$t = y + 1$$

$$x = 1 + 2(y + 1)$$

$$x = 1 + 2y + 2$$

$$x = 3 + 2y$$

t	0	$\frac{1}{2}$	1
x	1	2	3
y	-1	$-\frac{1}{2}$	0



Sketch the parametric curve (indicate with an arrow the direction in which the curve is traced as t is increases)

Then find the Cartesian

$$x = 1 + t, \quad y = t^2 - 4t, \quad 0 \leq t \leq 5$$

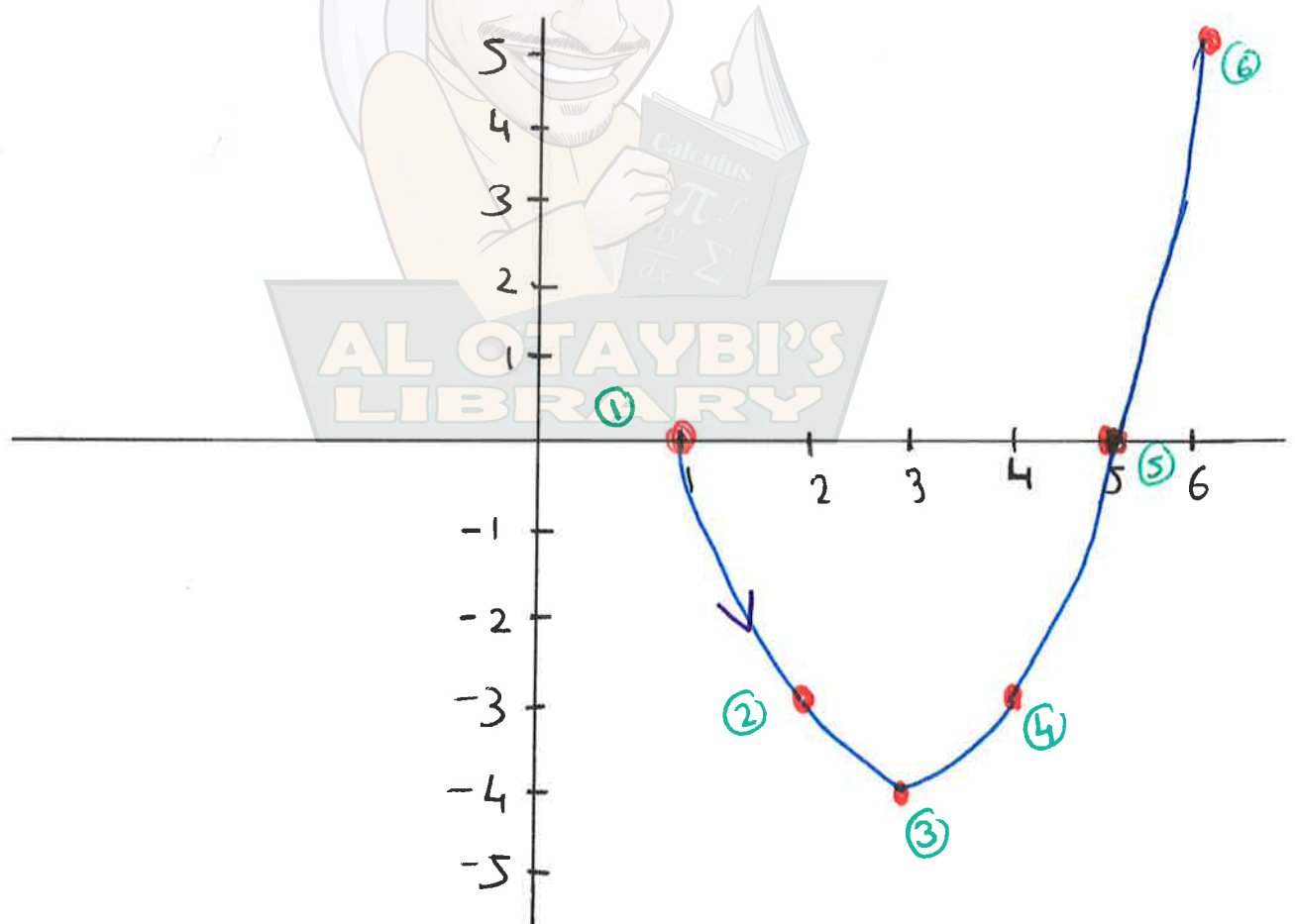
$$t = x - 1 \quad y = t^2 - 4t$$

$$y = (x-1)^2 - 4(x-1)$$

$$y = (x-1)[x-1-4]$$

$$y = (x-1)[x-5]$$

t	0	1	2	3	4	5
x	1	2	3	4	5	6
y	0	-3	-4	-3	0	5



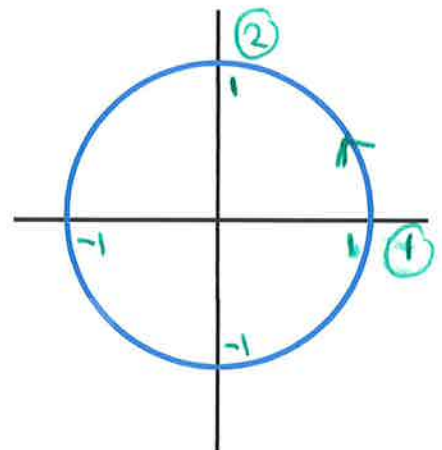
Find the equation in x and y and indicate the orientation of the parametric curve such that

$$x = \cos t, \quad y = \sin t \quad 0 \leq t \leq 2\pi$$

$$\cos^2 t + \sin^2 t = 1$$

$$x^2 + y^2 = 1$$

t	0	$\frac{\pi}{2}$
x	1	0
y	0	1



Find the equation in x and y and indicate the orientation of the parametric curve such that

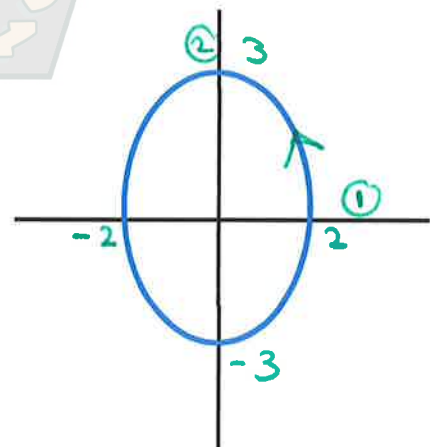
$$x = 2\cos t, \quad y = 3\sin t \quad 0 \leq t \leq 2\pi$$

$$\cos t = \frac{x}{2} \quad \sin t = \frac{y}{3}$$

$$\cos^2 t + \sin^2 t = 1$$

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$

t	0	$\frac{\pi}{2}$
x	2	0
y	0	3





Remark



The general form of the equation of circle

$$(x - a)^2 + (y - b)^2 = c$$

$$\text{Center} = (a, b), \quad \text{Radius} = \sqrt{c}$$

Find the center and radius of

$$(x + 2)^2 + (y - 5)^2 = 7$$

$$\text{Center} = (-2, 5) \quad \text{Radius} = \sqrt{7}$$



Tricky question



Show that the parametric equation

$$x = 1 + 2\cos t, \quad y = -3 + 2\sin t, \quad t \in [0, 2]$$

Defined as a circle with center $P(1, -3)$ and radius $r = 2$

$$x - 1 = 2 \cos t$$

$$y + 3 = 2 \sin t$$

$$\cos t = \frac{x-1}{2}$$

$$\sin t = \frac{y+3}{2}$$

$$\cos^2 t + \sin^2 t = 1$$

$$\left(\frac{x-1}{2}\right)^2 + \left(\frac{y+3}{2}\right)^2 = 1$$

$$\frac{(x-1)^2}{4} + \frac{(y+3)^2}{4} = 1$$

$$\times 4 \quad (x-1)^2 + (y+3)^2 = 4 \longrightarrow \text{Equation of circle}$$

$$\text{Center} = (1, -3) \quad \text{Radius} = \sqrt{4} = 2$$

Sketch the parametric curve (indicate with an arrow the direction in which the curve is traced as t is increases)

Then find the equation in x and y system

$$x = -2 + t^2, \quad y = 1 + 2t^2, \quad t \in \mathbb{R}$$

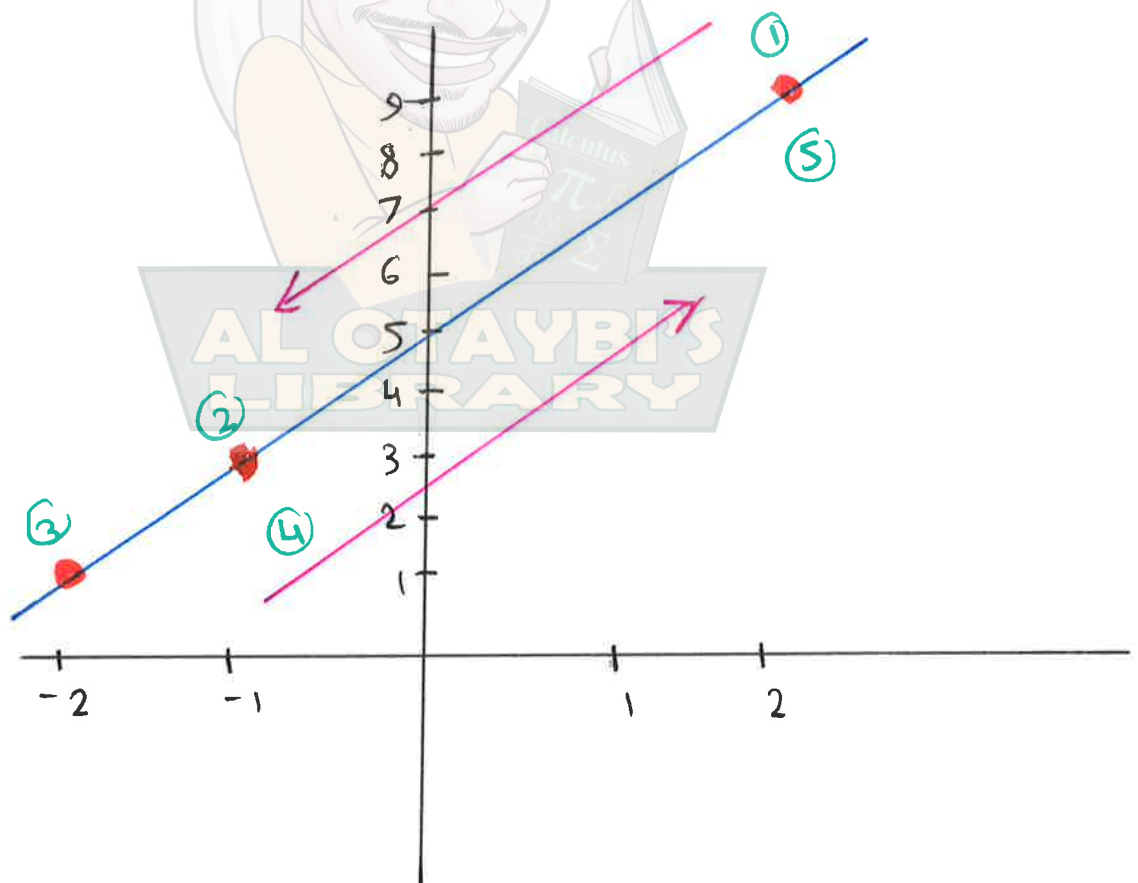
$$t^2 = x + 2 \quad y = 1 + 2t^2$$

$$y = 1 + 2(x + 2)$$

$$y = 1 + 2x + 4$$

$$y = 2x + 5$$

t	-2	-1	0	1	2
x	2	-1	-2	-1	2
y	9	3	1	3	9





Parametric curves



Arc length

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Surface area

about x - axis

$$S = 2\pi \int_a^b y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

about y - axis

$$S = 2\pi \int_a^b x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Arc
Length

5) Let C be the parametric curve

$$x = \cos^3 t, \quad y = \sin^3 t \quad \text{where } 0 \leq t \leq \frac{\pi}{2}$$

Find the length of C

Arc
Length

$$L = \int_0^{\frac{\pi}{2}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 3 \cos^2 t (-\sin t) = -3 \cos^2 t \sin t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = (-3 \cos^2 t \sin t)^2 = 9 \cos^4 t \sin^2 t$$

$$\frac{dy}{dt} = 3 \sin^2 t \cos t$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = (3 \sin^2 t \cos t)^2 = 9 \sin^4 t \cos^2 t$$

$$\begin{aligned} \Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= 9 \cos^4 t \sin^2 t + 9 \sin^4 t \cos^2 t \\ &= 9 \cos^2 t \sin^2 t [\cos^2 t + \sin^2 t] \\ &= 9 \cos^2 t \sin^2 t \quad (1) \end{aligned}$$

$$L = \int_0^{\frac{\pi}{2}} \sqrt{9 \cos^2 t \sin^2 t} dt$$

$$= 3 \int_0^{\frac{\pi}{2}} \cos t \sin t dt$$

$$u = \sin t \rightarrow du = \cos t dt$$

$$= 3 \int u du = 3 \frac{u^2}{2}$$

$$= \frac{3}{2} [\sin^2 t]_0^{\frac{\pi}{2}}$$

$$= \frac{3}{2} [1 - 0] = \frac{3}{2}$$

6) A curve C is given by the parametric equations $x = \sin t + \cos t$
 $y = \sin t - \cos t$ where $-\pi \leq t \leq \pi$
Find the length of the curve C

$$L = \int_{-\pi}^{\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = \cos t - \sin t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = (\cos t - \sin t)^2 = \cos^2 t + \sin^2 t - 2\cos t \sin t \\ = 1 - 2\cos t \sin t$$

$$\frac{dy}{dt} = \cos t + \sin t$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = (\cos t + \sin t)^2 = \cos^2 t + \sin^2 t + 2\cos t \sin t \\ = 1 + 2\cos t \sin t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 1 - 2\cos t \sin t + 1 + 2\cos t \sin t \\ = 2$$

$$L = \int_{-\pi}^{\pi} \sqrt{2} dt$$

$$= \sqrt{2} [t]_{-\pi}^{\pi}$$

$$= \sqrt{2} [\pi - (-\pi)]$$

$$= \sqrt{2} \cdot 2\pi$$

Consider the curve C

$$x = \frac{1}{2} \ln(1 - t^2) \quad , \quad y = \cos^{-1} t \quad 0 \leq t \leq \frac{3}{4}$$

Find the length of C

$$L = \int_0^{\frac{3}{4}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = \frac{1}{2} \frac{-2t}{1-t^2} = \frac{-t}{1-t^2}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = \left(\frac{-t}{1-t^2}\right)^2 = \frac{t^2}{(1-t^2)^2}$$

$$\frac{dy}{dt} = \frac{-1}{\sqrt{1-t^2}}$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(\frac{-1}{\sqrt{1-t^2}}\right)^2 = \frac{1}{1-t^2}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{t^2}{(1-t^2)^2} + \frac{1}{1-t^2}$$

$$= \frac{t^2}{(1-t^2)^2} + \frac{1-t^2}{(1-t^2)(1-t^2)}$$

$$= \frac{t^2 + 1 - t^2}{(1-t^2)^2} = \frac{1}{(1-t^2)^2}$$

$$L = \int_0^{\frac{3}{4}} \sqrt{\frac{1}{(1-t^2)^2}} dt = \int_0^{\frac{3}{4}} \frac{1}{1-t^2} dt$$

$$= \int_0^{\frac{3}{4}} \frac{1}{1-t^2} dt = - \int_0^{\frac{3}{4}} \frac{1}{t^2-1} dt$$

$$= -\frac{1}{2} \left[\ln \left| \frac{t-1}{t+1} \right| \right]_0^{\frac{3}{4}} = -\frac{1}{2} \left[\ln \left| \frac{\frac{3}{4}-1}{\frac{3}{4}+1} \right| - \ln 1 \right]$$

$$= -\frac{1}{2} \left[\ln \frac{1}{7} \right]$$

Sol #2 Partial Fraction

8) Find the arc length of the curve C whose parametric equation are
 $x = \ln \sqrt{1+t^2}$, $y = \tan^{-1}t$, $t \in [0, 1]$

$$L = \int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$x = \ln(1+t^2)^{\frac{1}{2}} = \frac{1}{2} \ln(1+t^2)$$

$$\frac{dx}{dt} = \frac{1}{2} \cdot \frac{2t}{1+t^2} = \frac{t}{1+t^2}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = \left(\frac{t}{1+t^2}\right)^2 = \frac{t^2}{(1+t^2)^2}$$

$$\frac{dy}{dt} = \frac{1}{t^2+1}$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(\frac{1}{t^2+1}\right)^2 = \frac{1}{(t^2+1)^2}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{t^2}{(1+t^2)^2} + \frac{1}{(t^2+1)^2}$$

$$= \frac{t^2+1}{(1+t^2)^2} = \frac{1}{1+t^2}$$

$$L = \int_0^1 \sqrt{\frac{1}{1+t^2}} dt$$

$$= \int_0^1 \frac{1}{\sqrt{1+t^2}} dt$$

Let $t = \tan \theta$
 $dt = \sec^2 \theta d\theta$



$$= \int \frac{1}{\sqrt{1+\tan^2 \theta}} \cdot \sec^2 \theta d\theta$$

$$= \int \frac{1}{\sec \theta} \sec^2 \theta d\theta = \ln |\sec \theta + \tan \theta|$$

$$= [\ln |\sqrt{t^2+1} + t|]_0^1 = [\ln |\sqrt{2}+1| - \ln |1|] = \ln |\sqrt{2}+1|$$

Set up the integral of arc length of the curve given by $x = 1 + e^{2t}$ and $y = 1 - e^t$ for $-\ln 2 \leq t \leq 0$

$$L = \int_{-\ln 2}^0 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 2e^{2t}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = (2e^{2t})^2 = 4e^{4t}$$

$$\frac{dy}{dt} = -e^t$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = (-e^t)^2 = e^{2t}$$

$$\begin{aligned} \Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= 4e^{4t} + e^{2t} \\ &= e^{2t} [4e^{2t} + 1] \end{aligned}$$

$$L = \int_{-\ln 2}^0 \sqrt{e^{2t} (4e^{2t} + 1)} dt$$

$$= \int_{-\ln 2}^0 e^t \sqrt{4e^{2t} + 1} dt$$

Let C be the curve with parametric equations

$$x(t) = \frac{1}{2}t^2, \quad y(t) = \frac{1}{3}(2t+1)^{3/2}, \quad 0 \leq t \leq 4.$$

Find the arc length.

Fall 2017

$$L = \int_0^4 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = t \longrightarrow \left(\frac{dx}{dt}\right)^2 = t^2$$

$$\frac{dy}{dt} = \frac{1}{3} \cdot \frac{3}{2} (2t+1)^{-1/2} \cdot 2 \longrightarrow \left(\frac{dy}{dt}\right)^2 = 2t+1$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = t^2 + 2t + 1$$

$$L = \int_0^4 \sqrt{t^2 + 2t + 1} dt$$

$$= \int_0^4 \sqrt{(t+1)^2} dt$$

$$= \int_0^4 (t+1) dt$$

$$= \left[\frac{t^2}{2} + t \right]_0^4$$

$$= \frac{16}{2} + 4 - 0$$

$$= 8 + 4 = 12$$

$$\begin{aligned} t^2 + 2t + 1 \\ (t+1)^2 - 1 + 1 \\ (t+1)^2 \end{aligned}$$

Let C be the curve with parametric equations :

Fall 2018

$$x = \frac{1}{3}t^3 + 1, \quad y = 2t^2 + 1$$

Find the length of the arc of C corresponding to $0 \leq t \leq 3$.

$$L = \int_0^3 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = t^2 \rightarrow \left(\frac{dx}{dt}\right)^2 = (t^2)^2 = t^4$$

$$\frac{dy}{dt} = 4t \rightarrow \left(\frac{dy}{dt}\right)^2 = (4t)^2 = 16t^2$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = t^4 + 16t^2 = t^2(t^2 + 16)$$

$$L = \int_0^3 \sqrt{t^2(t^2 + 16)} dt$$

$$= \int_0^3 t \sqrt{t^2 + 16} dt$$

$$\text{Let } u = t^2 + 16$$

$$du = 2t dt \rightarrow \frac{1}{2} du = t dt$$

$$= \frac{1}{2} \int \sqrt{u} du = \frac{1}{2} \cdot \frac{2}{3} u^{\frac{3}{2}}$$

$$= \frac{1}{3} \left[(t^2 + 16)^{\frac{3}{2}} \right]_0^3 = \frac{1}{3} \left[(25)^{\frac{3}{2}} - (16)^{\frac{3}{2}} \right]$$

$$= \frac{1}{3} [5^3 - 4^3]$$

Surface
Area

10) Find the surface area generated by rotating the parametric curve
 $x = \cos t + \sin t$, $y = \sin t - \cos t$, $0 \leq t \leq \frac{\pi}{2}$
around the y-axis.

Surface
Area

$$S = 2\pi \int_0^{\frac{\pi}{2}} x \cdot \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = -\sin t + \cos t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = (-\sin t + \cos t)^2 = \sin^2 t + \cos^2 t - 2\sin t \cos t \\ = 1 - 2\sin t \cos t$$

$$\frac{dy}{dt} = \cos t + \sin t$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = (\cos t + \sin t)^2 = \cos^2 t + \sin^2 t + 2\sin t \cos t \\ = 1 + 2\sin t \cos t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \cancel{1 - 2\sin t \cos t} + \cancel{1 + 2\sin t \cos t} \\ = 2$$

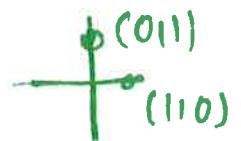
$$S = 2\pi \int_0^{\frac{\pi}{2}} (\cos t + \sin t) \sqrt{2} dt$$

$$= 2\pi \sqrt{2} [\sin t - \cos t]_0^{\frac{\pi}{2}}$$

$$= 2\sqrt{2} \pi [(1-0) - (0-1)]$$

$$= 2\sqrt{2} \pi [2]$$

$$= 4\sqrt{2} \pi$$



12) Let C be the plane curve parameterized by :

$$x = 2t - \sin 2t \quad y = \cos 2t, \quad 0 \leq t \leq \frac{\pi}{4}.$$

(a) Find the length of C.

(b) Find the area of the surface obtained by rotating C about the x axis.

$$(a) \quad L = \int_0^{\frac{\pi}{4}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 2 - 2 \cos 2t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = (2 - 2 \cos 2t)^2 = 4 - 8 \cos 2t + 4 \cos^2 2t$$

$$\frac{dy}{dt} = -2 \sin 2t$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = (-2 \sin 2t)^2 = 4 \sin^2 2t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$$

$$= 4 - 8 \cos 2t + 4 \cos^2 2t + 4 \sin^2 2t$$

$$= 4 - 8 \cos 2t + 4 = 8 - 8 \cos 2t = 8(1 - \cos 2t)$$

$$L = \int_0^{\frac{\pi}{4}} \sqrt{8(1 - \cos 2t)} dt$$

$$= \sqrt{8} \int_0^{\frac{\pi}{4}} \sqrt{1 - \cos 2t} dt$$

$$= \sqrt{8} \int_0^{\frac{\pi}{4}} \sqrt{2 \sin^2 t} dt$$

$$= \sqrt{8} \sqrt{2} \int_0^{\frac{\pi}{4}} \sin t dt$$

$$= \sqrt{16} [-\cos t]_0^{\frac{\pi}{4}} = 4 \left[-\frac{1}{\sqrt{2}} - 1\right]$$

$$\sin^2 t = \frac{1}{2}(1 - \cos 2t)$$
$$2 \sin^2 t = 1 - \cos 2t$$



ملاحظة

$$(b) \quad S = 2\pi \int_0^{\frac{\pi}{4}} y \cdot \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

From (a)

$$= 2\pi \int_0^{\frac{\pi}{4}} \cos 2t \cdot \sqrt{8(1 - \cos 2t)} dt$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \cos 2t \sqrt{8 \cdot 2 \sin^2 t} dt$$

$$= 2\pi \int_0^{\frac{\pi}{4}} \cos 2t \sqrt{16 \sin^2 t} dt$$

$$= 2\pi \cdot 4 \int_0^{\frac{\pi}{4}} \cos 2t \sin t dt$$

الرؤيا مختلفة

$$= 2\pi \cdot 4 \cdot \frac{1}{2} \int_0^{\frac{\pi}{4}} [\sin(t-2t) + \sin(t+2t)] dt$$

$$= 4\pi \int_0^{\frac{\pi}{4}} [-\sin(-t) + \sin 3t] dt$$

$$= 4\pi \left[\cos t - \frac{\cos 3t}{3} \right]_0^{\frac{\pi}{4}}$$

$$= 4\pi \left[\left(\frac{1}{\sqrt{2}} + \frac{1}{3} \right) - \left(1 - \frac{1}{3} \right) \right]$$

Let C be the curve with parametric equations

$$x = t - \ln t, \quad y = 4\sqrt{t}, \quad 1 \leq t \leq 4$$

a) Find the length of C .

b) Find the area of the surface obtained by rotating C about the x -axis.

Jan 2017

$$(a) \quad L = \int_1^4 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 1 - \frac{1}{t} = \frac{t-1}{t} \rightarrow \left(\frac{dx}{dt}\right)^2 = \left(\frac{t-1}{t}\right)^2 = \frac{t^2 - 2t + 1}{t^2}$$

$$\frac{dy}{dt} = 4 \cdot \frac{1}{2\sqrt{t}} = \frac{2}{\sqrt{t}} \rightarrow \left(\frac{dy}{dt}\right)^2 = \left(\frac{2}{\sqrt{t}}\right)^2 = \frac{4}{t}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{t^2 - 2t + 1}{t^2} + \frac{4}{t}$$
$$= \frac{t^2 - 2t + 1}{t^2} + \frac{4t}{t^2} = \frac{t^2 + 2t + 1}{t^2}$$

$$L = \int_1^4 \sqrt{\frac{t^2 + 2t + 1}{t^2}} dt$$

$$= \int_1^4 \frac{\sqrt{t^2 + 2t + 1}}{\sqrt{t^2}} dt \rightarrow \frac{t^2 + 2t + 1}{(t+1)^2 - 1 + 1}$$
$$= \int_1^4 \frac{\sqrt{(t+1)^2}}{t} dt$$

$$= \int_1^4 \left[\frac{t}{t} + \frac{1}{t} \right] dt = \int_1^4 \left[1 + \frac{1}{t} \right] dt$$

$$= \left[t + \ln|t| \right]_1^4 = (4 + \ln 4) - (1 + \ln 1) = \boxed{3 + \ln 4}$$

$$(b) \quad S = 2\pi \int_1^4 y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= 2\pi \int_1^4 4\sqrt{t} \sqrt{\frac{(t+1)^2}{t^2}} dt$$

$$= 8\pi \int_1^4 \sqrt{t} \frac{t+1}{t} dt$$

$$= 8\pi \int_1^4 \frac{t+1}{\sqrt{t}} dt$$

$$= 8\pi \int_1^4 \left(\frac{t}{\sqrt{t}} + \frac{1}{\sqrt{t}} \right) dt$$

$$= 8\pi \int_1^4 (t^{\frac{1}{2}} + t^{-\frac{1}{2}}) dt$$

$$= 8\pi \left[\frac{2}{3} t^{\frac{3}{2}} + 2t^{\frac{1}{2}} \right]_1^4$$

$$= 8\pi \left[\left(\frac{2}{3} (4)^{\frac{3}{2}} + 2(4)^{\frac{1}{2}} \right) - \left(\frac{2}{3} + 2 \right) \right]$$

$$x = e^{\frac{t}{2}} \cos t, \quad y = e^{\frac{t}{2}} \sin t, \quad 0 \leq t \leq 2\pi$$

(a) Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{5}{4}e^t$

(b) The curve C is revolved about the x-axis. Find the area of the resulting surface.

$$\begin{aligned} \text{(a)} \quad \frac{dx}{dt} &= e^{\frac{t}{2}} \cdot \frac{1}{2} \cos t + e^{\frac{t}{2}} (-\sin t) \\ &= \frac{1}{2} e^{\frac{t}{2}} \cos t - e^{\frac{t}{2}} \sin t \end{aligned}$$

$$\begin{aligned} \Rightarrow \left(\frac{dx}{dt}\right)^2 &= \left(\frac{1}{2} e^{\frac{t}{2}} \cos t - e^{\frac{t}{2}} \sin t\right)^2 \\ &= \frac{1}{4} e^t \cos^2 t - e^t \cos t \sin t + e^t \sin^2 t \end{aligned}$$

$$\frac{dy}{dt} = e^{\frac{t}{2}} \cdot \frac{1}{2} \sin t + e^{\frac{t}{2}} \cos t$$

$$\begin{aligned} \Rightarrow \left(\frac{dy}{dt}\right)^2 &= \left(\frac{1}{2} e^{\frac{t}{2}} \sin t + e^{\frac{t}{2}} \cos t\right)^2 \\ &= \frac{1}{4} e^t \sin^2 t + e^t \sin t \cos t + e^t \cos^2 t \end{aligned}$$

$$\begin{aligned} \Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= \frac{1}{4} e^t \cos^2 t - e^t \cos t \sin t + e^t \sin^2 t + \frac{1}{4} e^t \sin^2 t + e^t \sin t \cos t + e^t \cos^2 t \\ &= e^t \left[\frac{1}{4} \cos^2 t + \sin^2 t + \frac{1}{4} \sin^2 t + \cos^2 t \right] \\ &= e^t \left[\frac{1}{4} \cos^2 t + \frac{1}{4} \sin^2 t + \sin^2 t + \cos^2 t \right] \\ &= e^t \left[\frac{1}{4} (\cos^2 t + \sin^2 t) + 1 \right] \\ &= e^t \left[\frac{1}{4} (1) + 1 \right] = \frac{5}{4} e^t \end{aligned}$$

$\Rightarrow$ See Next page
Plz!

(b)

$$S = 2\pi \int_0^{2\pi} y \cdot \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= 2\pi \int_0^{2\pi} (e^{\frac{t}{2}} \sin t) \sqrt{\frac{\pi}{4}} e^t dt$$

$$= 2\pi \int_0^{2\pi} e^{\frac{t}{2}} \sin t \sqrt{\frac{\pi}{4}} \sqrt{e^t} dt$$

$$= 2\pi \sqrt{\frac{\pi}{4}} \int_0^{2\pi} e^{\frac{t}{2}} \sin t e^{\frac{t}{2}} dt$$

$$= 2\pi \frac{\sqrt{5}}{2} \int_0^{2\pi} e^t \sin t dt = I$$

$$\begin{aligned} u &= \sin t & dv &= e^t dt \\ du &= \cos t & v &= e^t \end{aligned}$$

تکامل
لولی

$$- \sin t e^t - \int e^t \cos t dt$$

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$$\begin{aligned} u &= \cos t & dv &= e^t dt \\ du &= -\sin t dt & v &= e^t \end{aligned}$$

$$\cos t e^t + \int e^t \sin t dt$$

$$I = -\sin t e^t - [\cos t e^t + I]$$

$$I = -\sin t e^t - \cos t e^t - I$$

$$2I = -\sin t e^t - \cos t e^t$$

$$I = \frac{1}{2} [-\sin t e^t - \cos t e^t]$$

$$\pi \sqrt{5} \frac{1}{2} [-\sin t e^t - \cos t e^t]_0^{2\pi}$$

$$\frac{1}{2} \pi \sqrt{5} [(0 - (11e^{2\pi})) - (0 - 1)]$$

$$\frac{1}{2} \pi \sqrt{5} [-e^{2\pi} + 1]$$



Area under parametric curve



$$A = \int_a^b y \, dx$$

Find the area enclosed by the x-axis and the curve

$$x = t^3, \quad y = t - t^2$$

$$x = t^3$$

$$y = t - t^2$$

$$y = 0$$

$$y = y$$

$$t - t^2 = 0$$

$$t(1-t) = 0$$

$$\begin{matrix} \nearrow t=0 \\ \searrow t=1 \end{matrix}$$

$$A = \int_0^1 y \, dx$$

$$= \int_0^1 (t - t^2) 3t^2 \, dt$$

$$= 3 \int_0^1 (t^3 - t^4) \, dt$$

$$= 3 \left[\frac{t^4}{4} - \frac{t^5}{5} \right]_0^1$$

$$= 3 \left[\frac{1}{4} - \frac{1}{5} - 0 \right]$$

$$= 3 \left[\frac{5-4}{20} \right] = \frac{3}{20}$$

Find the area of the region under the parametric curve

$$x(t) = t^4, \quad y(t) = \sqrt{t^2 + 1}, \quad 0 \leq t \leq 1.$$

Fall 2017

$$A = \int_0^1 y \, dx$$

$$= \int_0^1 \sqrt{t^2 + 1} \cdot 4t^3 \, dt$$

$$= 4 \int_0^1 t^3 \sqrt{t^2 + 1} \, dt$$

$$\text{Let } y = \sqrt{t^2 + 1}$$

$$y^2 = t^2 + 1 \rightarrow t^2 = y^2 - 1$$

$$2y = 2t \, dt$$

$$y = t \, dt$$

$$A = 4 \int_0^1 t^3 \sqrt{t^2 + 1} \, dt$$

$$= 4 \int (y^2 - 1) y \cdot y \, dy$$

$$= 4 \int (y^2 - 1) y^2 \, dy$$

$$= 4 \int (y^4 - y^2) \, dy$$

$$= 4 \left[\frac{y^5}{5} - \frac{y^3}{3} \right] = 4 \left[\frac{(\sqrt{t^2 + 1})^5}{5} - \frac{(\sqrt{t^2 + 1})^3}{3} \right] \Bigg|_0^1$$

$$= 4 \left[\left[\frac{(\sqrt{2})^5}{5} - \frac{(\sqrt{2})^3}{3} \right] - \left[\frac{1}{5} - \frac{1}{3} \right] \right]$$



Tricky question !!



Setup the integral of center of mass of the region above the x - axis and below

$$x = t - \sin t, \quad y = 1 - \cos t, \quad 0 \leq t \leq 2\pi$$

$$A = \int_0^{2\pi} [y_2 - y_1] dx$$

$\downarrow$ قيمة y $\downarrow$ 0 $\downarrow$ مشتقة x

$$= \int_0^{2\pi} (1 - \cos t) (1 - \cos t) dt$$

$$\bar{X} = \frac{1}{A} \int_0^{2\pi} x [y_2 - y_1] dx$$

$\downarrow$ مشتقة x $\downarrow$ قيمة y $\downarrow$ 0 $\downarrow$ مشتقة x

$$= \frac{1}{A} \int_0^{2\pi} (t - \sin t) (1 - \cos t) (1 - \cos t) dt$$

$$\bar{Y} = \frac{1}{2A} \int_0^{2\pi} [y_2^2 - y_1^2] dx$$

$\downarrow$ مشتقة x $\downarrow$ قيمة y $\downarrow$ 0

$$= \frac{1}{2A} \int_0^{2\pi} (1 - \cos t)^2 (1 - \cos t) dt$$

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi

* يمكن إيجار $\bar{X}$ من الرسمة *

The derivative of the parametric curves

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$



خطوات الحل



only when you want to find the equation of

ايجاد النقطة
 (x, y)

$$t = \dots \begin{cases} x = \dots \\ y = \dots \end{cases}$$

tangent

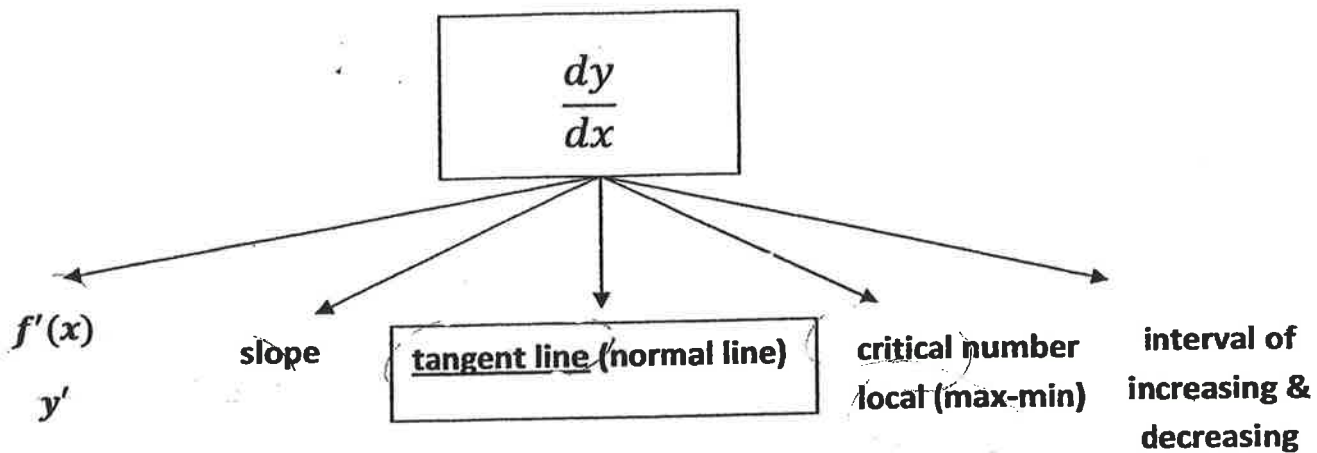
normal!

إذا عدي نقطة
أوضح فيهم

(1) ايجاد $\frac{dx}{dt}$

(2) ايجاد $\frac{dy}{dt}$

(3) قسمة 2 على 1 $\leftarrow \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$



Equation

$$y - y_0 = m(x - x_0)$$

$$m_{\text{normal}} = \frac{-1}{m_{\text{tangent}}}$$

Horizontal

$$\frac{dy}{dx} = 0$$

أصفار بسط المشتقة

Vertical

$$\frac{dy}{dx} \text{ undefind}$$

أصفار مقام المشتقة

Parallel to the line y

$$\frac{dy}{dx} = y'$$

Perpendicular to the line y

$$\frac{dy}{dx} = \frac{-1}{y'}$$

has slope

$$\frac{dy}{dx} = \text{slope}$$

A curve C has parameterization

$$x = 2t^3 - 6t, \quad y = 2t^3 + 3t^2, \quad -1 \leq t \leq 1$$

Find the coordinates of the points on C at which the tangent line has slope $\frac{1}{3}$

$$\textcircled{1} \quad \frac{dx}{dt} = 6t^2 - 6$$

$$\textcircled{2} \quad \frac{dy}{dt} = 6t^2 + 6t$$

$$\textcircled{3} \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{6t^2 + 6t}{6t^2 - 6} = \frac{\cancel{6}t(t+1)}{\cancel{6}(t^2-1)}$$
$$= \frac{\cancel{t}(t+1)}{(t-1)\cancel{(t+1)}} = \frac{t}{t-1}$$

$\Rightarrow \therefore$ tangent line of C has slope $= \frac{1}{3}$

$$\Rightarrow \therefore \frac{t}{t-1} = \frac{1}{3}$$

$$3t = t - 1$$

$$2t = -1 \longrightarrow t = -\frac{1}{2} \in [-1, 1]$$

للازم اطلع x و y
بالقويض

$$\text{at } t = -\frac{1}{2} \begin{cases} x = 2\left(-\frac{1}{8}\right) - 6\left(-\frac{1}{2}\right) \\ = -\frac{1}{4} + 3 = -\frac{1}{4} + \frac{12}{4} = \frac{11}{4} \\ y = 2\left(-\frac{1}{8}\right) + 3\left(\frac{1}{4}\right) \\ = -\frac{1}{4} + \frac{3}{4} = \frac{2}{4} = \frac{1}{2} \end{cases}$$

$$\Rightarrow \therefore P\left(\frac{11}{4}, \frac{1}{2}\right)$$

Consider the parametric equation

$$x = t \cos t, \quad y = t \sin t \quad \text{for } t \in [0, 2\pi]$$

Find the equation of the tangent line at $t = \frac{3\pi}{2}$

لأن المطلوب

equation
of tang. line

⇒ لازم أوجد

$$\text{at } t = \frac{3\pi}{2} \begin{cases} x = \frac{3\pi}{2}(0) = 0 \\ y = \frac{3\pi}{2}(-1) = -\frac{3\pi}{2} \end{cases}$$

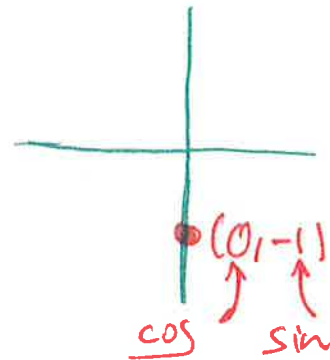
$$\Rightarrow P(0, -\frac{3\pi}{2})$$

① $\frac{dx}{dt} = (1)\cos t - t \sin t$

لأن المطلوب
نقطة
أوجد

$$\text{at } t = \frac{3\pi}{2}$$

$$\frac{dx}{dt} = 0 - \frac{3\pi}{2}(-1) = \frac{3\pi}{2}$$



② $\frac{dy}{dt} = (1)\sin t + t \cdot \cos t$

$$\text{at } t = \frac{3\pi}{2}$$

$$\frac{dy}{dt} = (-1) + 0 = -1$$

③ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-1}{\frac{3\pi}{2}} = -\frac{2}{3\pi}$

∴ equation of the tangent line

$$y - y_0 = m(x - x_0)$$

$$y - (-\frac{3\pi}{2}) = -\frac{2}{3\pi}(x - 0)$$

Extra

Equation of
normal line

$$y - (-\frac{3\pi}{2}) = \frac{3\pi}{2}(x - 0)$$



Tricky question !!



Let C be the curve given by the parametric equation

$$x = \frac{1}{3}t^3, \quad y = \frac{1}{2}t^2, \quad 0 \leq t \leq 1$$

Find the equation of the tangent line at $P\left(\frac{1}{24}, \frac{1}{8}\right)$

$$x = \frac{1}{24} \rightarrow \frac{1}{24} = \frac{1}{3}t^3$$

$$\times 3 \quad t^3 = \frac{1}{8} \rightarrow t = \frac{1}{2}$$

$$y = \frac{1}{8} \rightarrow \frac{1}{8} = \frac{1}{2}t^2$$

$$\times 2 \quad t^2 = \frac{1}{4} \rightarrow t = \frac{1}{2} \text{ or } t = -\frac{1}{2}$$

$$\Rightarrow \therefore t = \frac{1}{2}$$

تَمُّ الْفِتْنَةِ
الْمَشْرُوكَةِ

$$\frac{dx}{dt} = t^2$$

$$\rightarrow \text{at } t = \frac{1}{2} \rightarrow \frac{dx}{dt} = \frac{1}{4}$$

$$\frac{dy}{dt} = t$$

$$\rightarrow \text{at } t = \frac{1}{2} \rightarrow \frac{dy}{dt} = \frac{1}{2}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{1}{2}}{\frac{1}{4}} = \frac{1}{\frac{1}{2}} = 2$$

$\therefore$ The equation of the tangent line

$$y - y_0 = m(x - x_0)$$

$$y - \frac{1}{8} = 2\left(x - \frac{1}{24}\right)$$

Let C be the curve given by the parametric equation

$$x = \tanh(t) , \quad y = \sqrt{t+4} , \quad 0 \leq t \leq 4$$

Find the slope of the tangent line at $t = 0$

$$\frac{dx}{dt} = \text{sech}^2 t \quad \xrightarrow{\text{at } t=0} \quad \frac{dx}{dt} = \text{sech}^2 0 = 1$$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t+4}} \quad \xrightarrow{\text{at } t=0} \quad \frac{dy}{dt} = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{1}{4}}{1} = \frac{1}{4}$$

$\therefore$ The slope of the tangent line = $\frac{1}{4}$

Find the slope of the tangent line to the parametric curve :

$$x(t) = e^{t-1} , \quad y(t) = \tan^{-1} t$$

at the point where $t = 1$.

Fall 2017

$$\frac{dx}{dt} = e^{t-1} \quad \xrightarrow{\text{at } t=1} \quad \frac{dx}{dt} = e^0 = 1$$

$$\frac{dy}{dt} = \frac{1}{t^2+1} \quad \xrightarrow{\text{at } t=1} \quad \frac{dy}{dt} = \frac{1}{2}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{1}{2}}{1} = \frac{1}{2}$$

$\therefore$ The slope of the tangent line = $\frac{1}{2}$

Let C be the curve given by the parametric equation

$$x = t^2 - 2t, \quad y = t^3, \quad t \in \mathbb{R}$$

Find the point on C at which the tangent line is :

- a) Horizontal b) Vertical

$$\frac{dx}{dt} = 2t - 2$$

$$\frac{dy}{dt} = 3t^2$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2}{2t-2}$$

a) $\therefore$ tangent line is horizontal $\Rightarrow \therefore \frac{dy}{dx} = 0$

$$\Rightarrow 3t^2 = 0 \Rightarrow t = 0 \in \mathbb{R}$$

$$\therefore \text{at } t=0 \begin{cases} x = 0 \\ y = 0 \end{cases} \Rightarrow (0, 0)$$

$$2t-2 \neq 0$$

b) $\therefore$ tangent line is vertical $\Rightarrow \therefore \frac{dy}{dx}$ undefined

$$\Rightarrow 2t-2 = 0 \Rightarrow t = 1 \in \mathbb{R}$$

$$\therefore \text{at } t=1 \begin{cases} x = 1-2 = -1 \\ y = (1)^3 = 1 \end{cases} \Rightarrow (-1, 1)$$

$$3t^2 \neq 0$$



Multiple choice



The curve with parametric equations

Jan 2017

$$x = \frac{1}{3}t^3 - 2t^2 + 4t, \quad y = t^2 + 4t, \quad t \in \mathbb{R}$$

has a vertical tangent line at :

- (a) $t = 2$ (b) $t = -2$ (c) $t = \pm 2$ (d) $t = 3$ (e) $t = 4$

$$\frac{dx}{dt} = t^2 - 4t + 4$$

$$\frac{dy}{dt} = 2t + 4$$

$$\frac{dy}{dx} = \frac{2t+4}{t^2-4t+4} = \frac{2(t+2)}{(t-2)(t-2)} = \frac{2(t+2)}{(t-2)^2}$$

$\therefore$ The curve has vertical tangent line :
 $\frac{dy}{dx}$ undefined

$$t-2 = 0$$

$$t = 2 \quad \text{(a)}$$



Multiple choice



The curve with parametric equations $x = t^3 - \frac{3}{2}t^2$, $y = e^t - 2t$

has a vertical tangent line at :

Fall 2018

- (a) $t = 0, 2$ (b) $t = 1, 2$ (c) $t = 0, 1$ (d) None of the above.

$$\frac{dx}{dt} = 3t^2 - 3t$$

$$\frac{dy}{dt} = e^t - 2$$

$$\frac{dy}{dx} = \frac{e^t - 2}{3t^2 - 3t}$$

The curve has vertical tangent line:

$\frac{dy}{dx}$ undefined

$$3t^2 - 3t = 0$$

$$3t(t-1) = 0$$

$$3t = 0 \quad t - 1 = 0$$

$$t = 0$$

$$t = 1$$

(c)

Let C is given by the parametric equations

$$x = \sin t + \cos t, \quad y = \sin t - \cos t \quad -\pi \leq t \leq \pi$$

Find the points $P(x, y)$ on C where it has :

a) horizontal tangent line.

b) Vertical tangent line.

$$\frac{dx}{dt} = \cos t - \sin t$$

$$\frac{dy}{dt} = \cos t + \sin t$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\cos t + \sin t}{\cos t - \sin t}$$

$$\Rightarrow \therefore \text{Horizontal tangent line} \Rightarrow \therefore \frac{dy}{dx} = 0$$

$$\therefore \cos t + \sin t = 0$$

$$\div \cos t \Rightarrow \text{للتبسيط}$$

$$1 + \tan t = 0$$

$$\tan t = -1$$

شئو
اللي تقطبي
1

$$\cos t - \sin t \neq 0$$

$$t = \frac{\pi}{4}$$

$$\Rightarrow t = \pi - \frac{\pi}{4} = \frac{3\pi}{4} \in [-\pi, \pi]$$

$$\Rightarrow t = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4} \notin [-\pi, \pi]$$

$$t = \frac{7\pi}{4} - 2\pi = -\frac{\pi}{4} \in [-\pi, \pi]$$

#	S	A
t		C #

$$\text{at } t = \frac{3\pi}{4} \begin{cases} x = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0 \\ y = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} \end{cases} \Rightarrow (0, \frac{2}{\sqrt{2}})$$

$$\text{at } t = -\frac{\pi}{4} \begin{cases} x = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = 0 \\ y = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{2}{\sqrt{2}} \end{cases} \Rightarrow (0, -\frac{2}{\sqrt{2}})$$

(b) ∴ Vertical tangent Line $\Rightarrow \frac{dy}{dx}$ DNE

$$\therefore \cos t - \sin t = 0$$

$$\div \cos t$$

$$1 - \tan t = 0$$

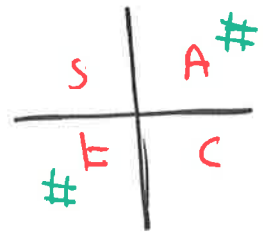
$$1 = \tan t$$

$$t = \frac{\pi}{4} \in (-\pi, \pi)$$

فصلك $t = \pi + \frac{\pi}{4} = \frac{5\pi}{4} \notin (-\pi, \pi)$

$$\cos t + \sin t \neq 0$$

$$t = \frac{\pi}{4}$$



$$t = \frac{5\pi}{4} - 2\pi = -\frac{3\pi}{4} \in (-\pi, \pi)$$

at $t = \frac{\pi}{4} \rightarrow x = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} \Rightarrow (\frac{2}{\sqrt{2}}, 0)$
 $y = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$

at $t = -\frac{3\pi}{4} \rightarrow x = \frac{-1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{2}{\sqrt{2}} \Rightarrow (-\frac{2}{\sqrt{2}}, 0)$
 $y = \frac{-1}{\sqrt{2}} - (\frac{-1}{\sqrt{2}}) = 0$

$$\Rightarrow (\frac{2}{\sqrt{2}}, 0), (-\frac{2}{\sqrt{2}}, 0)$$

Let C be the parametric curve

$$x = \cos^3 t, \quad y = \sin^3 t \quad 0 \leq t \leq \frac{\pi}{2}$$

Find the point(s) on C where the tangent line is parallel to the line $y = -\sqrt{3}x$

$$\frac{dx}{dt} = 3\cos^2 t (-\sin t) = -3\cos^2 t \sin t$$

$$\frac{dy}{dt} = 3\sin^2 t (\cos t)$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3\sin^2 t \cos t}{-3\cos^2 t \sin t} = \frac{\sin t}{-\cos t} = -\tan t$$

$$y = -\sqrt{3}x$$

$$y' = -\sqrt{3}$$

$\therefore$ the tangent line of C is parallel to y

$$\therefore \frac{dy}{dx} = y'$$

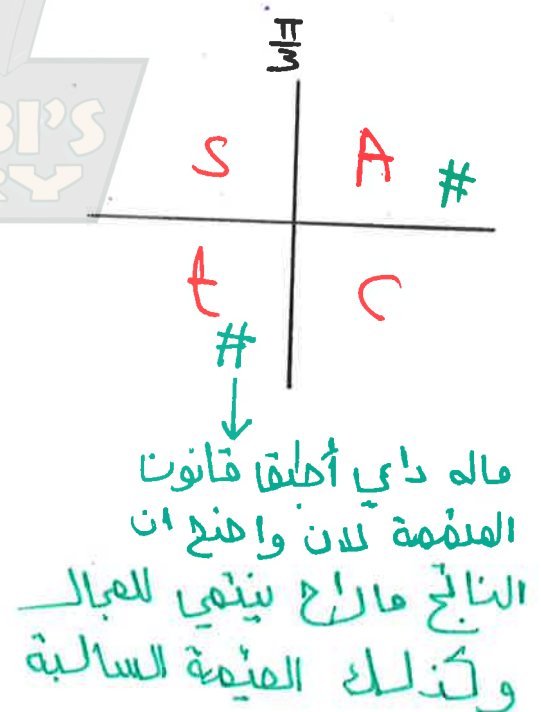
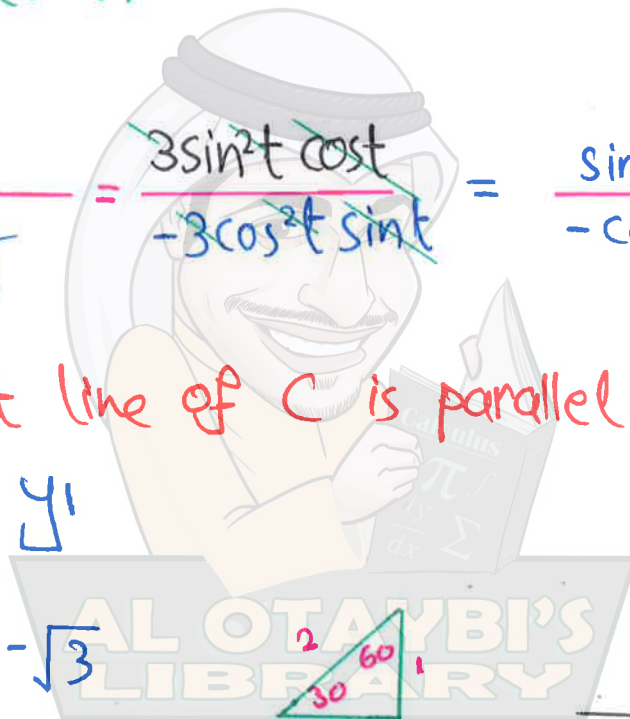
$$-\tan t = -\sqrt{3}$$

$$\tan t = \sqrt{3}$$

$$t = \frac{\pi}{3} \rightarrow x = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$y = \left(\frac{\sqrt{3}}{2}\right)^3$$

$$\Rightarrow P\left(\frac{1}{8}, \left(\frac{\sqrt{3}}{2}\right)^3\right)$$



Let C be the curve with parametric equations

May 2017

$$x = 2t^3, \quad y = 3t^2 - 5.$$

a) Find the equation for the tangent line to C at the point $(16, 7)$.

b) Find the length of the arc of C corresponding to $0 \leq t \leq \sqrt{3}$.

(a) at $(16, 7)$ $\begin{cases} 16 = 2t^3 \rightarrow 8 = t^3 \rightarrow t = 2 \\ 7 = 3t^2 - 5 \rightarrow 12 = 3t^2 \rightarrow t^2 = 4 \rightarrow t = \pm 2 \end{cases}$

$\Rightarrow \therefore \boxed{t=2}$ $P(16, 7)$

$$\frac{dx}{dt} = 6t^2 \xrightarrow{\text{at } t=2} \frac{dx}{dt} = 6(4) = 24$$

$$\frac{dy}{dt} = 6t \xrightarrow{\text{at } t=2} \frac{dy}{dt} = 6(2) = 12$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{12}{24} = \frac{1}{2}$$

$\therefore$ The equation of the tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 7 = \frac{1}{2}(x - 16)$$

$$(b) \quad L = \int_0^{\sqrt{3}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 6t^2 \longrightarrow \left(\frac{dx}{dt}\right)^2 = (6t^2)^2 = 36t^4$$

$$\frac{dy}{dt} = 6t \longrightarrow \left(\frac{dy}{dt}\right)^2 = (6t)^2 = 36t^2$$

$$\begin{aligned} \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= 36t^4 + 36t^2 \\ &= 36t^2 [t^2 + 1] \end{aligned}$$

$$L = \int_0^{\sqrt{3}} \sqrt{36t^2 [t^2 + 1]} dt$$

$$= \int_0^{\sqrt{3}} 6t \sqrt{t^2 + 1} dt$$

$$\text{Let } u = t^2 + 1$$

$$du = 2t dt \longrightarrow \frac{1}{2} du = t dt$$

$$= 6 \cdot \frac{1}{2} \int u^{\frac{1}{2}} du = \cancel{3} \frac{2}{\cancel{3}} u^{\frac{3}{2}}$$

$$= 2 \left[(t^2 + 1)^{\frac{3}{2}} \right]_0^{\sqrt{3}}$$

$$= 2 \left[(4)^{\frac{3}{2}} - 1^{\frac{3}{2}} \right]$$

$$= 2 [2^3 - 1] = 2 [8 - 1] = 14$$

$$\frac{d^2y}{dx^2} = \frac{\left(\frac{dy}{dt}\right)'}{\frac{dx}{dt}}$$

$$\frac{d^2y}{dx^2}$$

$$f''(x)$$

$$y''$$

Inflection point

نقطة الانعطاف

interval of concavity
(up - down)

فترات التقعر

خطوات الحل

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(1) إيجاد $\frac{dx}{dt}$

(2) إيجاد $\frac{dy}{dt}$

(3) قسمة 2 على 1 $\leftarrow \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

(4) اشتقاق 3 $\leftarrow \left(\frac{dy}{dx}\right)'$

(5) قسمة 4 على 1 $\leftarrow \frac{d^2y}{dx^2} = \frac{\left(\frac{dy}{dx}\right)'}{\frac{dx}{dt}}$

Suppose a curve is given by the parametric equation

$$x = t - \tan^{-1}t, \quad y = \sqrt{1+t^2}, \quad t \in [0, 1]$$

a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$

b) Find the arc length.

$$\textcircled{1} \quad \frac{dx}{dt} = 1 - \frac{1}{1+t^2} = \frac{1+t^2}{1+t^2} - \frac{1}{1+t^2} = \frac{t^2}{1+t^2}$$

$$\textcircled{2} \quad \frac{dy}{dt} = \frac{2t}{2\sqrt{1+t^2}} = \frac{t}{\sqrt{1+t^2}}$$

$$\begin{aligned} \textcircled{3} \quad \frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{t}{\sqrt{1+t^2}}}{\frac{t^2}{1+t^2}} \div \frac{t^2}{1+t^2} \\ &= \frac{t}{\sqrt{1+t^2}} \times \frac{1+t^2}{t^2} \\ &= \frac{1}{\sqrt{1+t^2}} \times \frac{\sqrt{1+t^2} \sqrt{1+t^2}}{t} = \frac{\sqrt{1+t^2}}{t} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad \left(\frac{dy}{dx} \right)' &= \left(\frac{\sqrt{1+t^2}}{t} \right)' = \frac{t \cdot \frac{2t}{2\sqrt{1+t^2}} - \sqrt{1+t^2} (1)}{t^2} \\ &= \frac{\frac{t^2}{\sqrt{1+t^2}} - \sqrt{1+t^2}}{t^2} \end{aligned}$$

$$\begin{aligned} \textcircled{5} \quad \frac{d^2y}{dx^2} &= \frac{\frac{t^2}{\sqrt{1+t^2}} - \sqrt{1+t^2}}{t^2} \div \frac{t^2}{1+t^2} \\ &= \frac{\frac{t^2}{\sqrt{1+t^2}} - \sqrt{1+t^2}}{t^2} \times \frac{1+t^2}{t^2} \end{aligned}$$

$$(b) \quad L = \int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\frac{dx}{dt} = 1 - \frac{1}{1+t^2} = \frac{1+t^2}{1+t^2} - \frac{1}{1+t^2} = \frac{t^2}{1+t^2}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 = \left(\frac{t^2}{1+t^2}\right)^2 = \frac{t^4}{(1+t^2)^2}$$

$$\frac{dy}{dt} = \frac{2t}{2\sqrt{1+t^2}} = \frac{t}{\sqrt{1+t^2}}$$

$$\Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(\frac{t}{\sqrt{1+t^2}}\right)^2 = \frac{t^2}{1+t^2}$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{t^4}{(1+t^2)^2} + \frac{t^2}{1+t^2}$$

$$= \frac{t^4}{(1+t^2)^2} + \frac{t^2(1+t^2)}{(1+t^2)(1+t^2)} = \frac{t^4 + t^2 + t^4}{(1+t^2)^2} = \frac{2t^4 + t^2}{(1+t^2)^2}$$

$$= \frac{t^2(2t^2+1)}{(1+t^2)^2}$$

$$L = \int_0^1 \sqrt{\frac{t^2(2t^2+1)}{(1+t^2)^2}} dt$$

$$= \int_0^1 \frac{t \sqrt{2t^2+1}}{1+t^2} dt$$

ملاحظة

$$y = \sqrt{2t^2+1}$$

$$y^2 = 2t^2+1$$

$$2y dy = 4t dt$$

الخط التالي ..
← تم حل هذا التفاضل 7.5

Let C be the curve given by the parametric equation

$$x = t - \sin t, \quad y = 1 - \cos t, \quad 0 \leq t \leq \frac{\pi}{2}$$

Find $\frac{d^2y}{dx^2}$

$$(1) \quad \frac{dx}{dt} = 1 - \cos t$$

$$(2) \quad \frac{dy}{dt} = \sin t$$

$$(3) \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\sin t}{1 - \cos t}$$

$$(4) \quad \left(\frac{dy}{dx}\right)' = \frac{(1 - \cos t) \cos t - \sin t \sin t}{(1 - \cos t)^2}$$

بفضل التبسيط

$$= \frac{\cos t - \cos^2 t - \sin^2 t}{(1 - \cos t)^2} = \frac{\cos t - (\cos^2 t + \sin^2 t)}{(1 - \cos t)^2}$$
$$= \frac{\cos t - 1}{(1 - \cos t)^2} = \frac{-\cancel{(1 - \cos t)}}{(1 - \cos t)^2} = \frac{-1}{1 - \cos t}$$

$$(5) \quad \frac{d^2y}{dx^2} = \frac{-1}{1 - \cos t} \div (1 - \cos t)$$

$$= \frac{-1}{1 - \cos t} * \frac{1}{1 - \cos t}$$

$$= \frac{-1}{(1 - \cos t)^2}$$

A curve C is given parametrically by

$$x = (t^2 + 1)e^t, \quad y = (t^3 + 2t^2)e^t$$

a) Find the points where the tangent line is vertical

b) Determine whether C is concave upwards or downwards at $t = 0$

$$\begin{aligned}\frac{dx}{dt} &= 2te^t + (t^2 + 1)e^t \\ &= e^t(2t + t^2 + 1) = e^t(t^2 + 2t + 1)\end{aligned}$$

$$\begin{aligned}\frac{dy}{dt} &= (3t^2 + 4t)e^t + (t^3 + 2t^2)e^t \\ &= e^t(3t^2 + 4t + t^3 + 2t^2) \\ &= e^t(t^3 + 5t^2 + 4t) = e^t \cdot t(t^2 + 5t + 4)\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{e^t t(t^2 + 5t + 4)}{e^t(t^2 + 2t + 1)} = \frac{t(t+1)(t+4)}{(t+1)(t+1)} \\ &= \frac{t^2 + 4t}{t+1}\end{aligned}$$

$\Rightarrow \therefore$ tangent line is Vertical

$$\begin{aligned}\therefore \frac{dy}{dx} \text{ undefined} &\Rightarrow t+1=0 \\ &t=-1 \\ &\begin{cases} x = (2)e^{-1} = \frac{2}{e} \\ y = (-1+2)e^{-1} = \frac{1}{e} \end{cases} \\ &\Rightarrow P\left(\frac{2}{e}, \frac{1}{e}\right)\end{aligned}$$

$$t^2 + 4t \neq 0$$

$$b) \left(\frac{dy}{dx} \right)' = \left(\frac{t^2 + 4t}{t+1} \right)'$$

$$= \frac{(t+1)(2t+4) - (t^2+4t)(1)}{(t+1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{(t+1)(2t+4) - (t^2+4t)}{(t+1)^2} \div e^t(t^2+2t+1)$$

$$= \frac{(t+1)(2t+4) - (t^2+4t)}{(t+1)^2} * \frac{1}{e^t(t^2+2t+1)}$$

at $t=0$

$$\frac{d^2y}{dx^2} = \frac{(1)(4) - 0}{(1)^2} * \frac{1}{(1)(1)}$$

$$= \frac{4}{1} * 1 = 4 > 0$$

$\Rightarrow \therefore$ at $t=0$

the curve is concave upward.

A curve C is given parametrically by

$$x = \frac{1}{2}t^2 + 1, \quad y = \frac{1}{3}t^3 - t, \quad t \in \mathbb{R}$$

a) Find the intervals of concavity

b) Find the inflection point(s) if any.

$$\textcircled{1} \quad \frac{dx}{dt} = t$$

$$\textcircled{2} \quad \frac{dy}{dt} = t^2 - 1$$

$$\textcircled{3} \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{t^2 - 1}{t}$$

$$\textcircled{4} \quad \left(\frac{dy}{dx} \right)' = \frac{t(2t) - (t^2 - 1)(1)}{t^2} = \frac{2t^2 - t^2 + 1}{t^2}$$

$$\begin{aligned} \textcircled{5} \quad \frac{d^2y}{dx^2} &= \frac{\frac{2t^2 - t^2 + 1}{t^2}}{t} \\ &= \frac{t^2 + 1}{t^2} \times \frac{1}{t} \\ &= \frac{t^2 + 1}{t^3} \end{aligned}$$

$\Rightarrow$ See Next Page p12!

$$\frac{d^2y}{dx^2} = \frac{t^2 + 1}{t^3}$$

$$= 0$$

$$t^2 + 1 = 0$$

$$t^2 = -1$$

No Ne

DNE

$$t = 0$$



$\frac{t^2 + 1}{t^3}$	$(-\infty, 0)$	$(0, \infty)$
	$\oplus$	$\oplus$
	$\ominus$	$\oplus$
	$\ominus$	$\oplus$
	$\cap$	$\cup$

Interval at which f is concave upward $(0, \infty)$

Interval at which f is concave downward $(-\infty, 0)$

الحل المطلوب الثاني ادوب وين الدالة صاعد نازل او العكس

at $t = 0$ $\begin{cases} x = 0 + 1 = 1 \\ y = 0 - 0 = 0 \end{cases}$

$\Rightarrow (1, 0)$ is the inflection point.

Let C be the curve given by the parametric equation

$$x = \frac{2}{3}(1-t)^{\frac{3}{2}}, \quad y = 2(1+t)^{\frac{1}{2}}, \quad 0 \leq t \leq 1$$

a) Find $\frac{d^2y}{dx^2}$

b) Is the curve C concave upward or downward on the given interval?

$$\frac{dx}{dt} = \frac{2}{3} \cdot \frac{3}{2} (1-t)^{\frac{1}{2}} (-1) = -\sqrt{1-t}$$

$$\frac{dy}{dt} = 2 \cdot \frac{1}{2} (1+t)^{-\frac{1}{2}} = \frac{1}{\sqrt{1+t}}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{1}{\sqrt{1+t}}}{-\sqrt{1-t}} = \frac{1}{-\sqrt{1-t} \cdot \sqrt{1+t}}$$

$$\left(\frac{dy}{dx}\right)' = \left(\frac{-1}{\sqrt{1-t^2}}\right)' = \frac{-1}{\sqrt{1-t^2}}$$

$$= \frac{0 - (-1) \cdot \frac{-1}{2} \cdot \frac{1}{\sqrt{1-t^2}}}{1-t^2} = \frac{-t}{1-t^2}$$

$$= \frac{-t}{(1-t^2)\sqrt{1-t^2}} = \frac{-t}{(1-t^2)^{\frac{3}{2}}}$$

$$\frac{d^2y}{dx^2} = \frac{-t}{(1-t^2)^{\frac{3}{2}}} \div (-\sqrt{1-t})$$

$$\frac{d^2y}{dx^2} = \frac{-t}{(1-t^2)^{\frac{3}{2}}} * \frac{-1}{\sqrt{1-t}}$$

$$= \frac{t}{(1-t^2)^{\frac{3}{2}} \sqrt{1-t}}$$

$$= 0$$

$$t=0 \notin (0,1)$$

البراءة هي الفترة
مفتوحة



DNE (undefined)

$$(1-t^2)^{\frac{3}{2}} = 0 \quad \sqrt{1-t} = 0$$

$$\downarrow$$

$$t^2 = 1$$

$$t=1$$

$$\notin (0,1)$$

$$t = \pm 1 \notin (0,1)$$



	(0, 1)
$\frac{t}{(1-t^2)^{\frac{3}{2}} \sqrt{1-t}}$	+
	+
	+

+

U

$\Rightarrow \therefore \frac{d^2y}{dx^2}$ is concave upward
on $(0,1)$